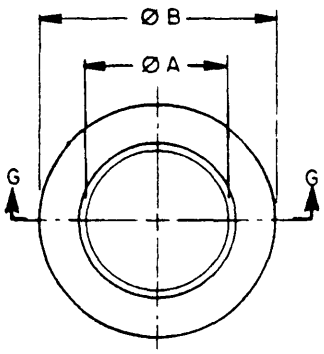
MILITARY SPECIFICATION SHEET

WASHER, CONCAVE AND WASHER CONVEX (SPHERICAL SET), STEEL, METRIC

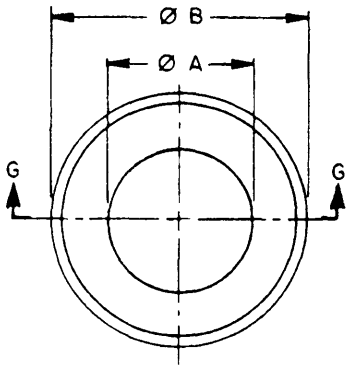
This specification is approved for use by all Departments and Agencies of the Department of Defense.

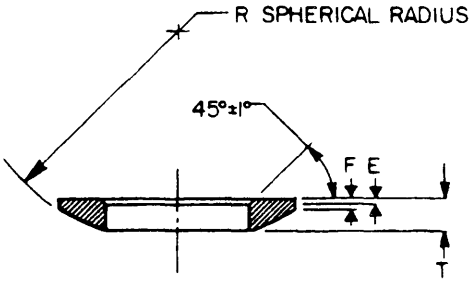
The requirements for acquiring the washers described herein shall consist of this specification and the latest issue of DOD-W-63546.

WASHER, CONVEX

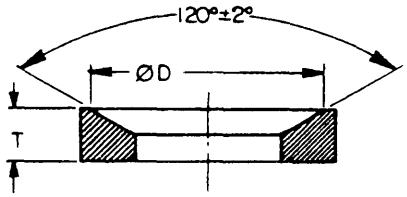


WASHER, CONCAVE





SECTION G-G



SECTION G-G

NOTES:

1. All dimensions are in millimeters.
2. Remove all burrs and sharp edges.

FIGURE 1. WASHER, CONCAVE AND WASHER CONVEX (SPHERICAL SET), STEEL, METRIC.

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REQUIREMENTS:

1. Material: Steel, carbon, composition 1010 to 1020 (UNS G10100 thru UNS G10200).

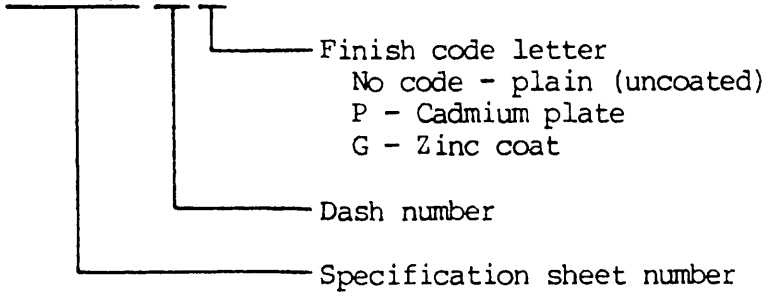
2. Heat treatment: Case harden total case depth 0.2 mm minimum.

3. Hardness: Surface hardness shall be 88 HR15N maximum, in accordance with ASTM E18.

4. Protective coating and finish. Plain (uncoated) or cadmium plated in accordance with QQ-P-416, Type II, Class 2 or zinc coated in accordance with ASTM B633, Type II, Fe/Zn 13 (see Requirement 5 herein).

5. Part number: The part number shall consist of the letter "D", the basic number of the specification sheet, a dash number from Table I or Table II and finish code letter as applicable.

Example: D63546/8-01 P



D63546/8-01P Indicates: Washer, convex, steel, metric, 6.4 mm I.D., and 12 mm OD., cadmium plate.

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TABLE I. DASH NUMBERS AND DIMENSIONS FOR CONVEX.

DASH NUMBER	ØA ID	TOL	ØB OD	TOL	E	TOL	F	TOL	T THICKNESS	TOL	R SPHERICAL RADIUS	TOL
01	6.4	+.15 0	12	0 -.18	0.3	±.01	0.7	±.05	2.3	±0.3	9	±.5
02	8.4	+.15 0	17	0 -.18	0.5		0.6	±.05	3.2		12	
03	10.5	+.18 0	21	0 -.21	0.5		0.8	±.05	4		15	
04	13	+.18 0	24	0 -.21	0.5		1.1	±.10	4.6		17	
05	17	+.18 0	30	0 -.25	0.5		1.3	±.20	5.3		22	
06	21	+.21 0	36	0 -.25	0.5		2		6.3		27	
07	25	+.21 0	44	0 -.25	0.8	±.05	2.4		8.2		32	
08	31	+.25 0	56	0 -.30	0.8		3.6		11.2		41	
09	37	+.25 0	68	0 -.30	1		4.6		14		50	
10	43	+.25 0	78	0 -.30	1		6.5		17		58	
11	50	+.25 0	92	0 -.35	1		8		21		67	

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TABLE II. DASH NUMBERS AND DIMENSIONS FOR CONCAVE.

DASH NUMBER	ØA ID		ØB OD		T THICKNESS		D	
		TOL		TOL		TOL		TOL
21	7.1	+.15 0	12	0 -.18	2.8		11	+.18 0
22	9.6	+.15 0	17	0 -.18	3.5		14.5	+.18 0
23	12	+.18 0	21	0 -.21	4.2		18.5	+.21 0
24	14.2	+.18 0	24	0 -.21	5		20	+.21 0
25	19	+.21 0	30	-.25	6.2	±0.3	26	+.21 0
26	23.2	+.21 0	36	0 -.25	7.5		31	+.25 0
27	28	+.21 0	44	0 -.25	9.5		37	+.25 0
28	35	+.25 0	56	0 -.30	12		49	+.25 0
29	42	+.25 0	68	0 -.30	15		60	+.30 0
30	49	+.25 0	78	0 -.30	18		70	+.30 0
31	56	+.30 0	92	0 -.35	22		82	+.35 0

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MATING ASSEMBLY

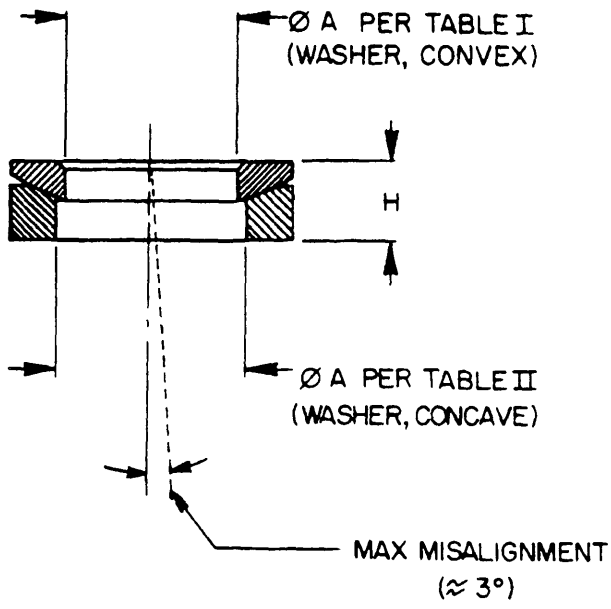


FIGURE 2. MATING ASSEMBLY.

TABLE III. MATING DIMENSIONS.

DASH NUMBER Ø A CONVEX	DASH NUMBER Ø A CONCAVE	H ASSEMBLY HEIGHT FOR MISALIGNMENT 0° (REF)
01	21	4
02	22	5
03	23	6.3
04	24	8
05	25	9.3
06	26	11.5
07	27	15
08	28	19.7
09	29	23
10	30	29
11	31	36

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Custodians:
Army - AR
Navy - OS

Review activities:
Army - AV
Navy - SH
DLA - IS
NSA - NS

User activities:
Army - AT
Navy - MC

Preparing activity:
Army - AR

Agent:
DLA- IS

(Project 5310-1435)

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